Point-of-Interest Recommender Systems

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Outline

- Data
- Social Networks
- Location-Based Social Networks
- Information Overload Problem
- Recommender Systems
- Point-of-Interest Recommender Systems
- Challenges
- References

Data

Social Networks

- Social Networks impact on Data Generation

Location-Based Social Networks

yelp

Gowalla

FOURSQUARE

Facebook Places

Who. What. When. And now **where.**

brightkite

people. places. friends.

Information Overload Problem

- Which items are better for costumer?
- Effective decision making

Recommender Systems

User Feedbacks

ratings, check-ins, buys, like

User or Object Profiles

attributes

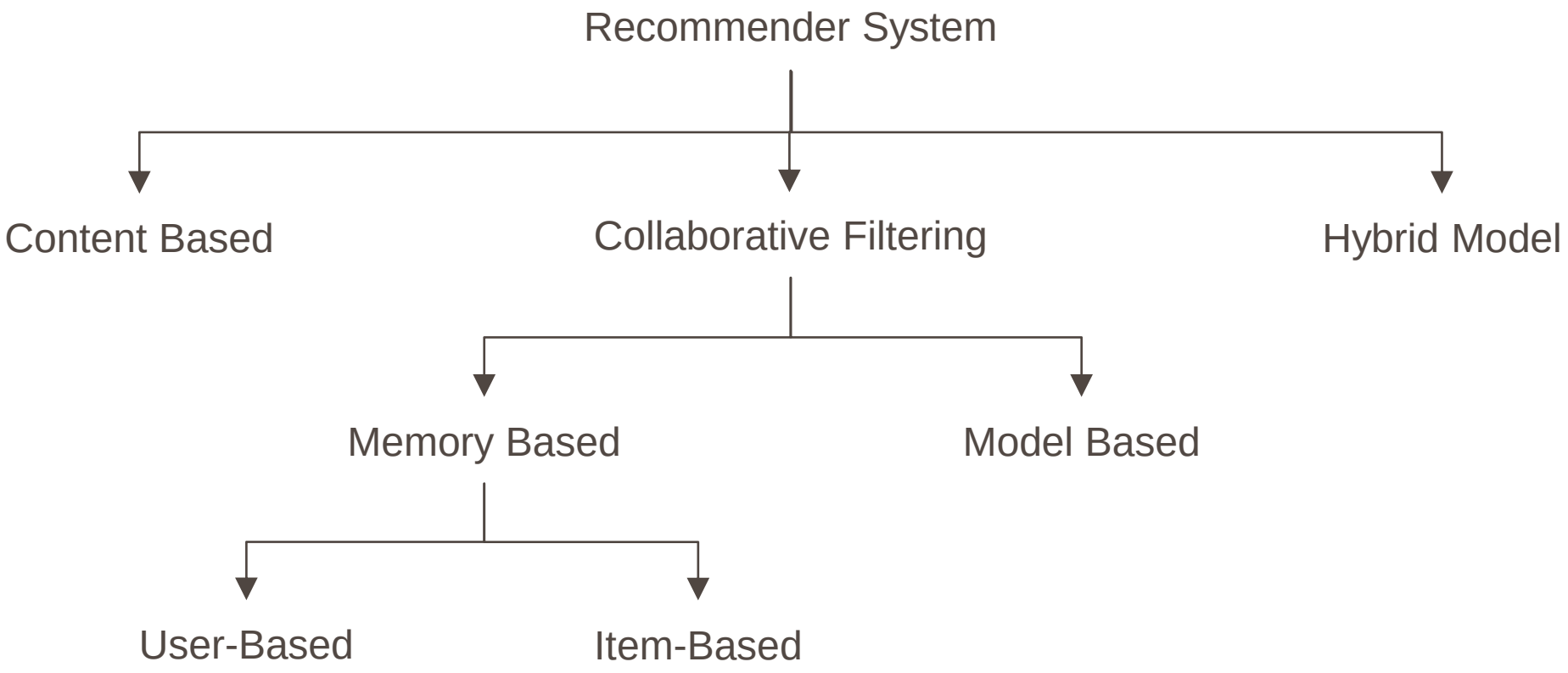
Context Information

location, time, friends, category, content,

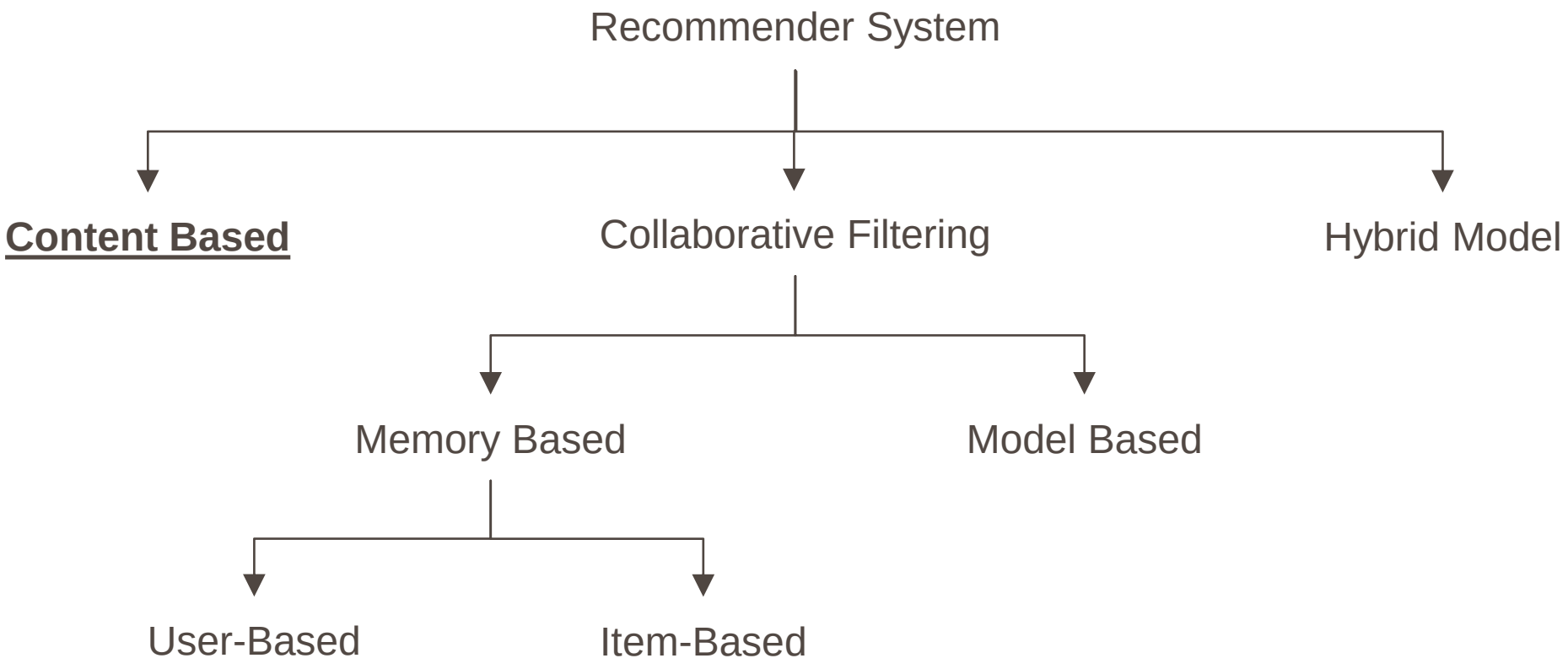
Recommender System

Recommended List

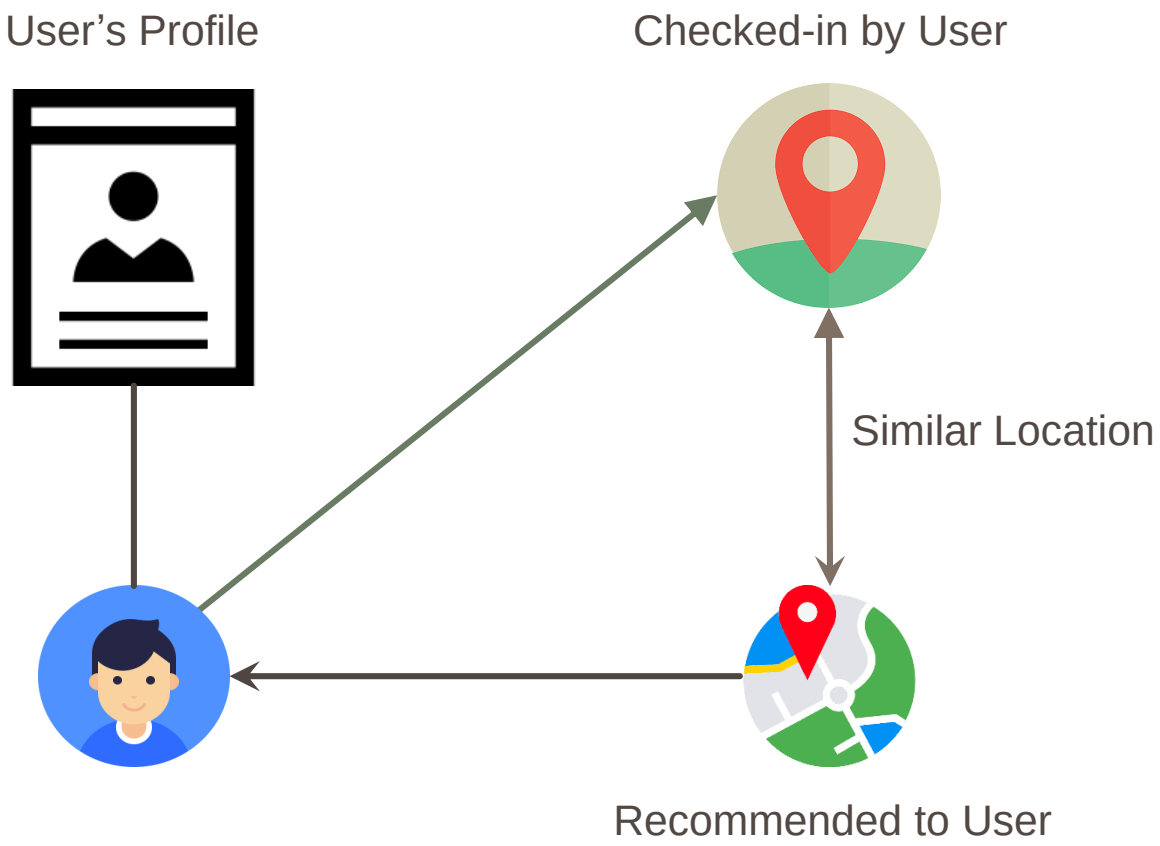
Recommendation Methods



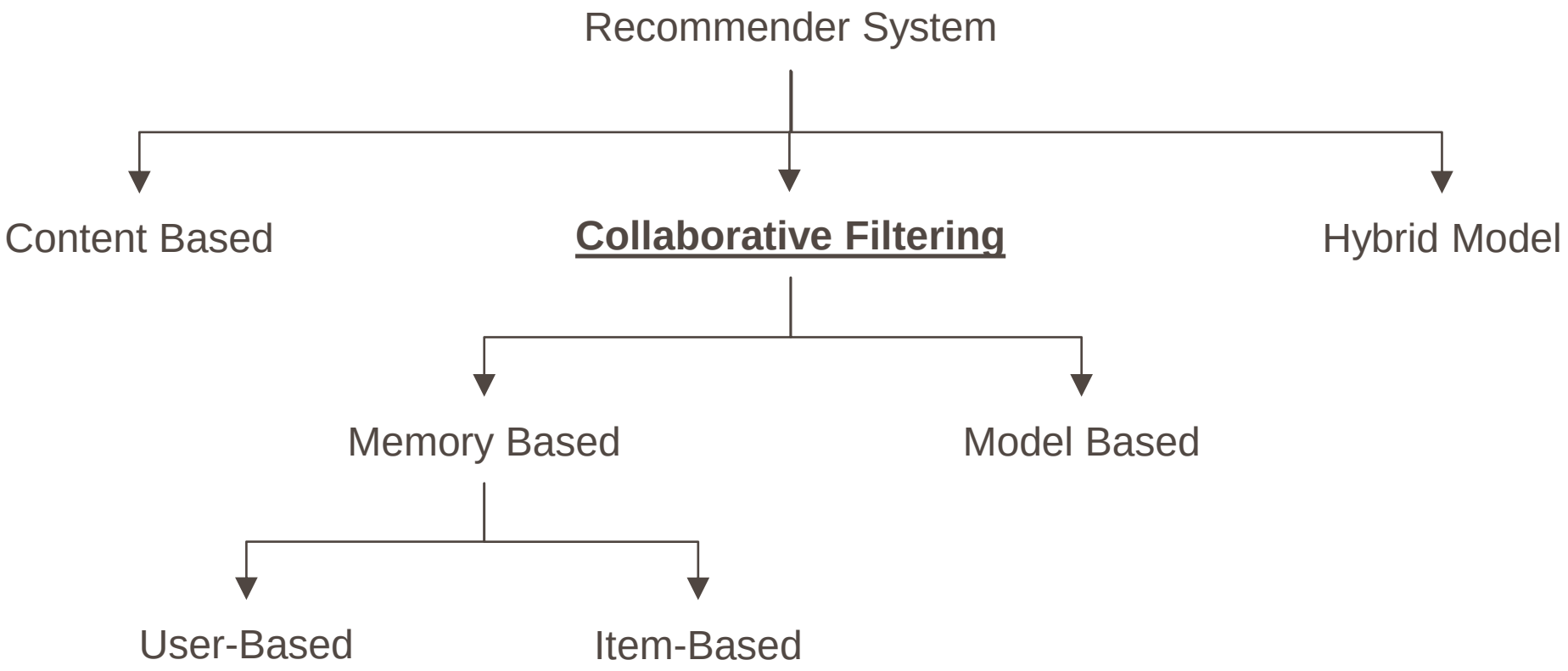
Recommendation Methods



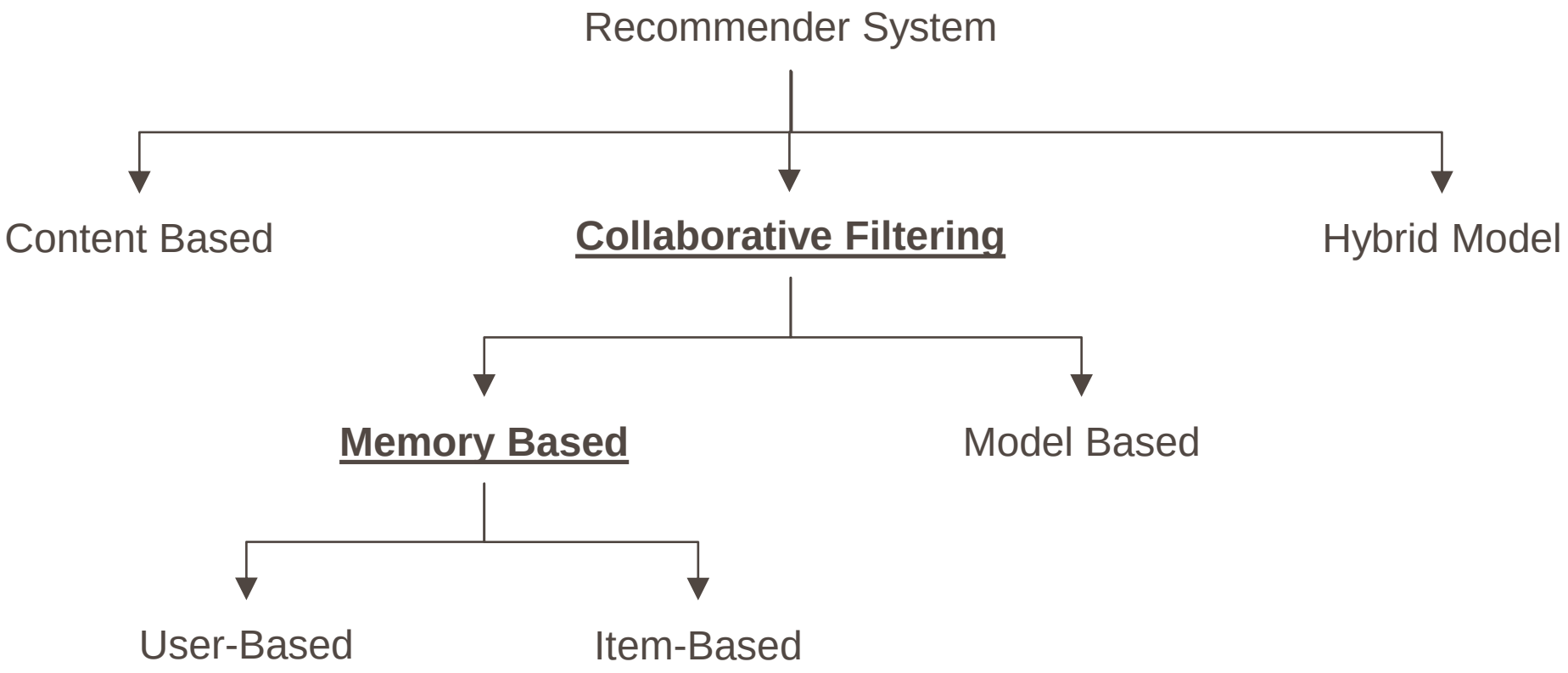
Content Based



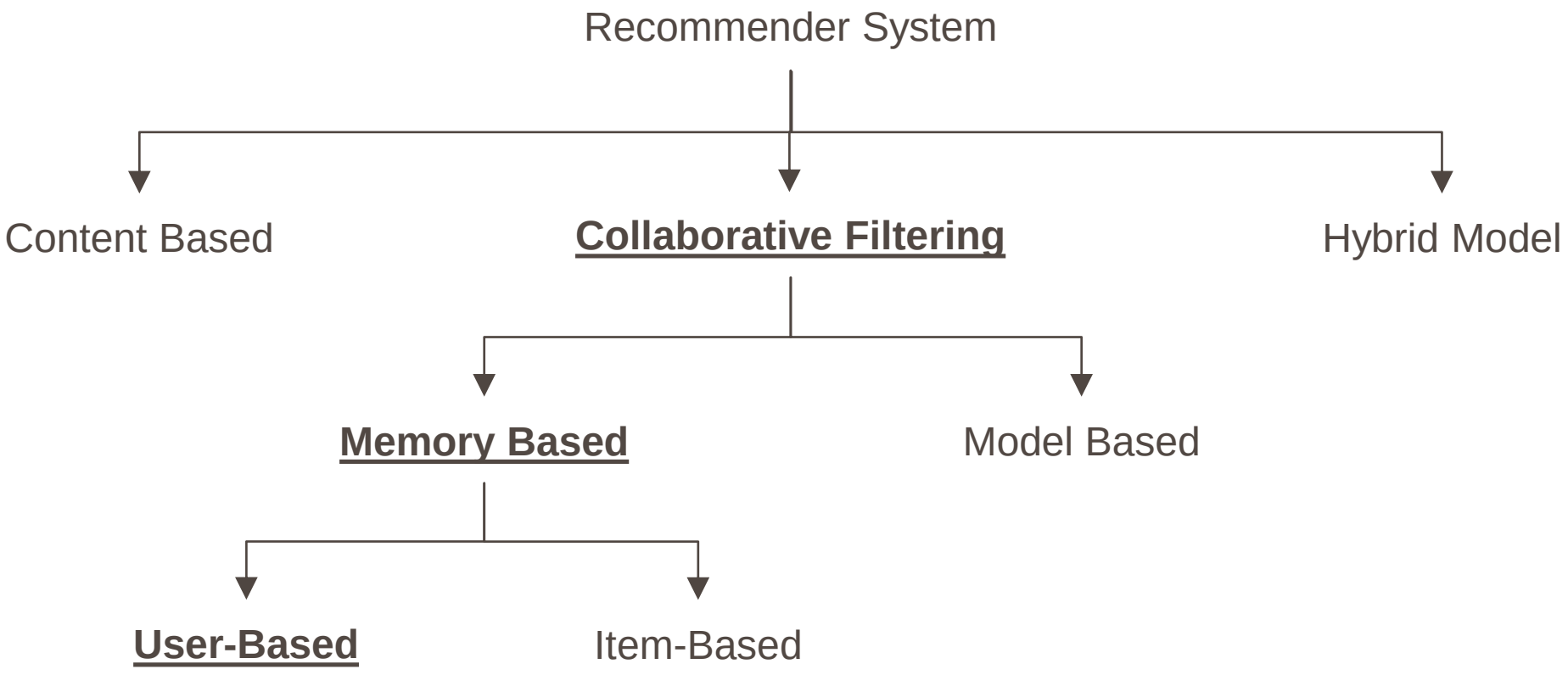
Recommendation Methods



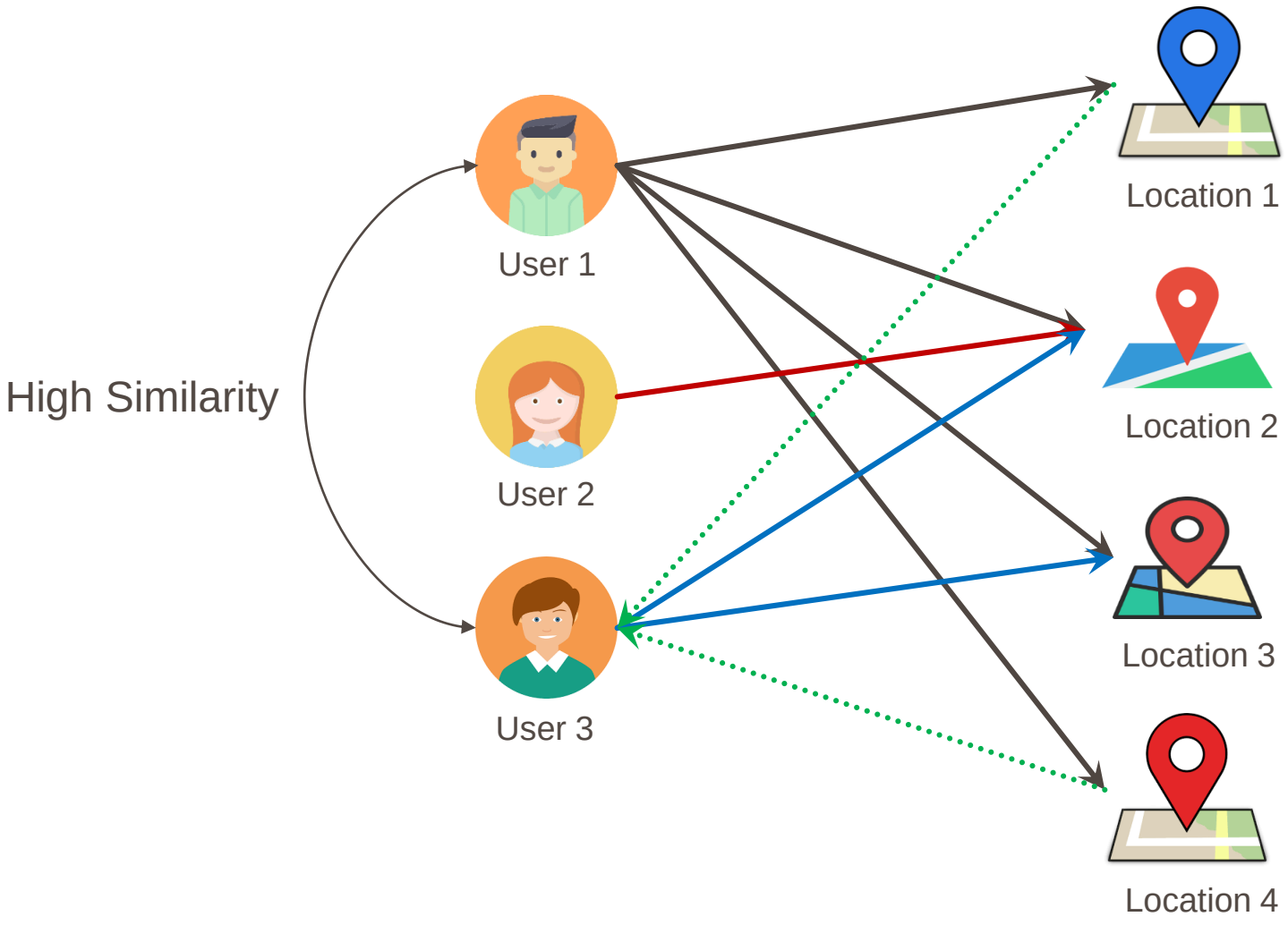
Recommendation Methods



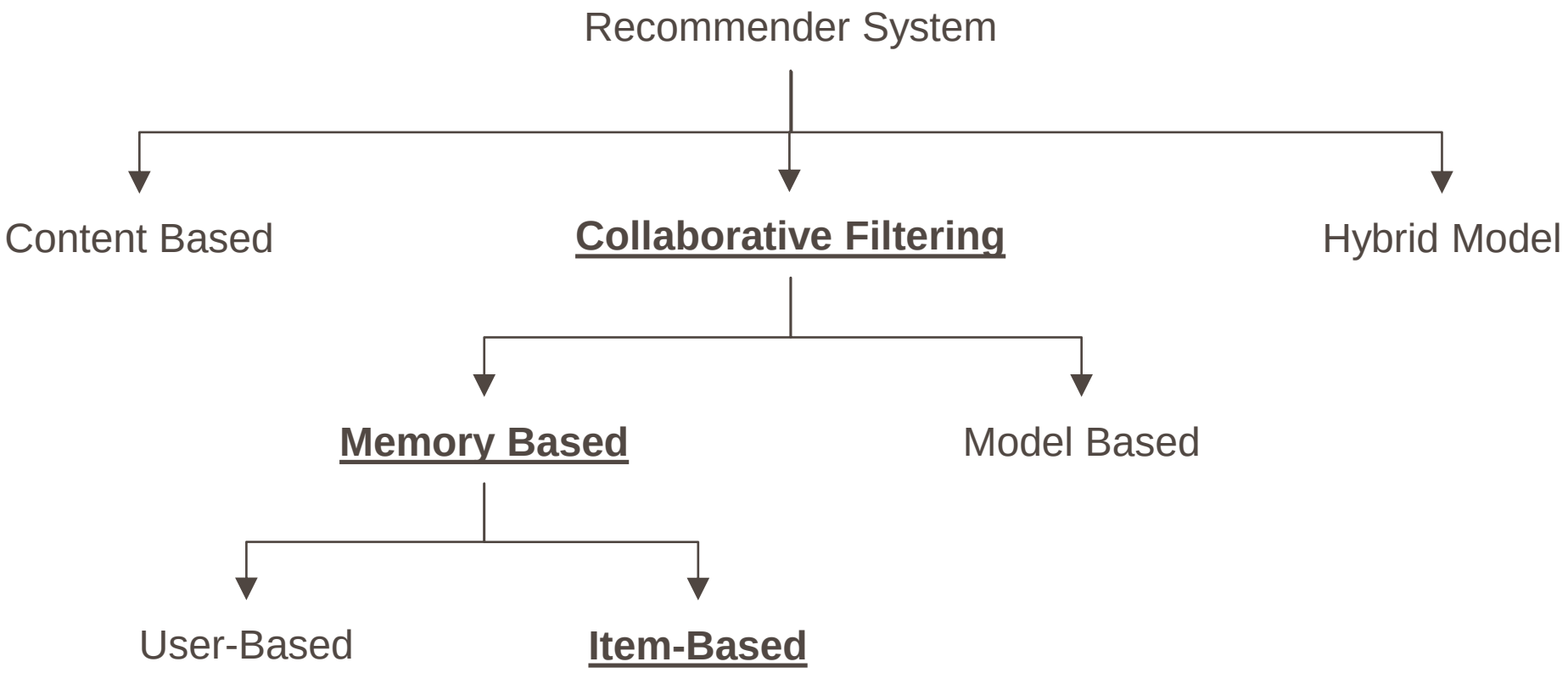
Recommendation Methods



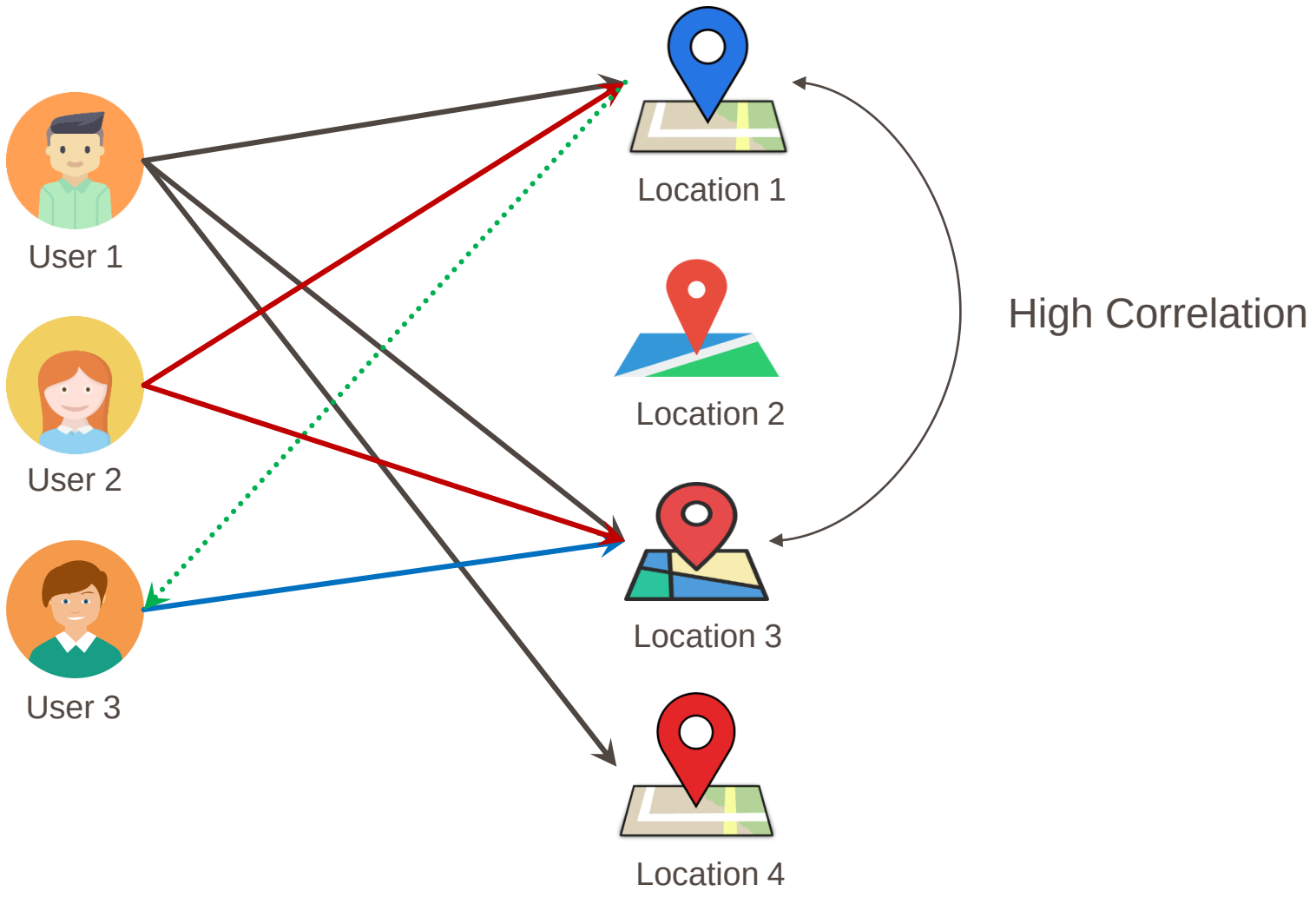
User-Based Collaborative Filtering



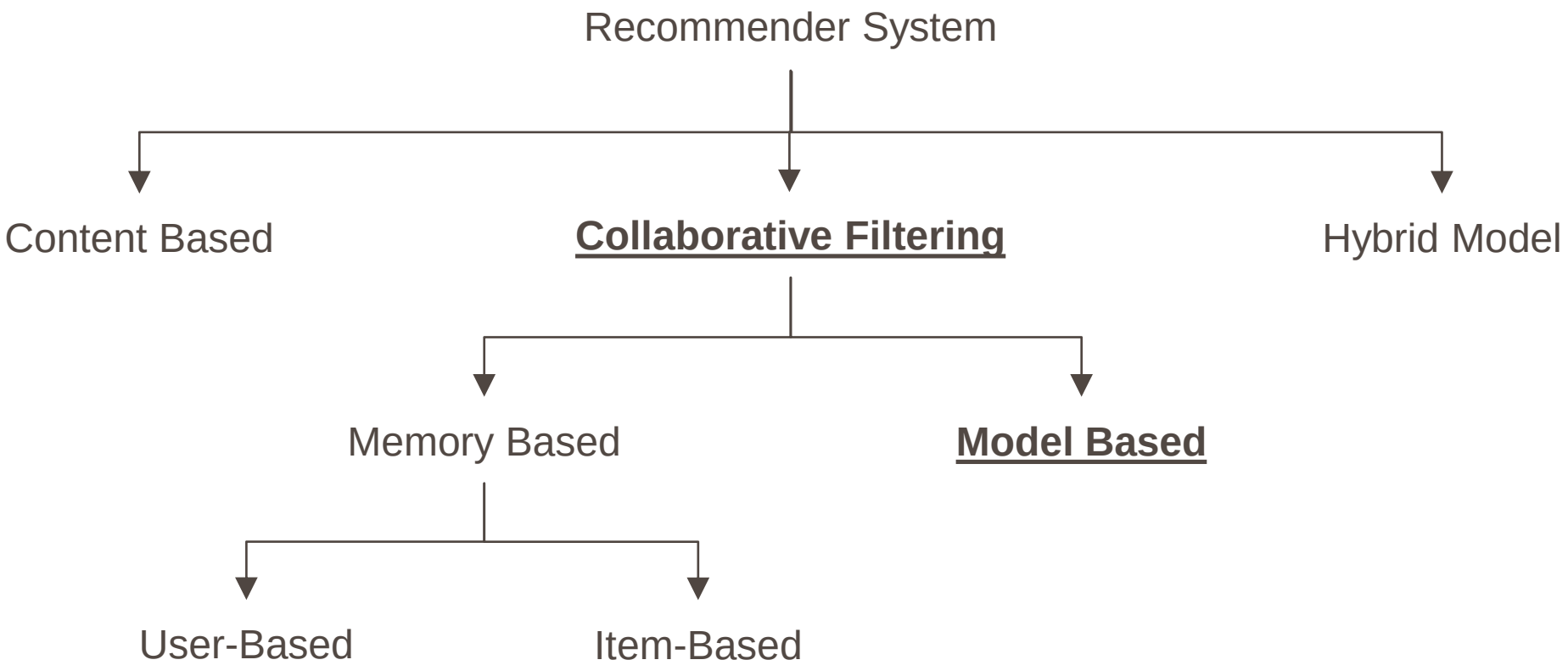
Recommendation Methods



Item-Based Collaborative Filtering

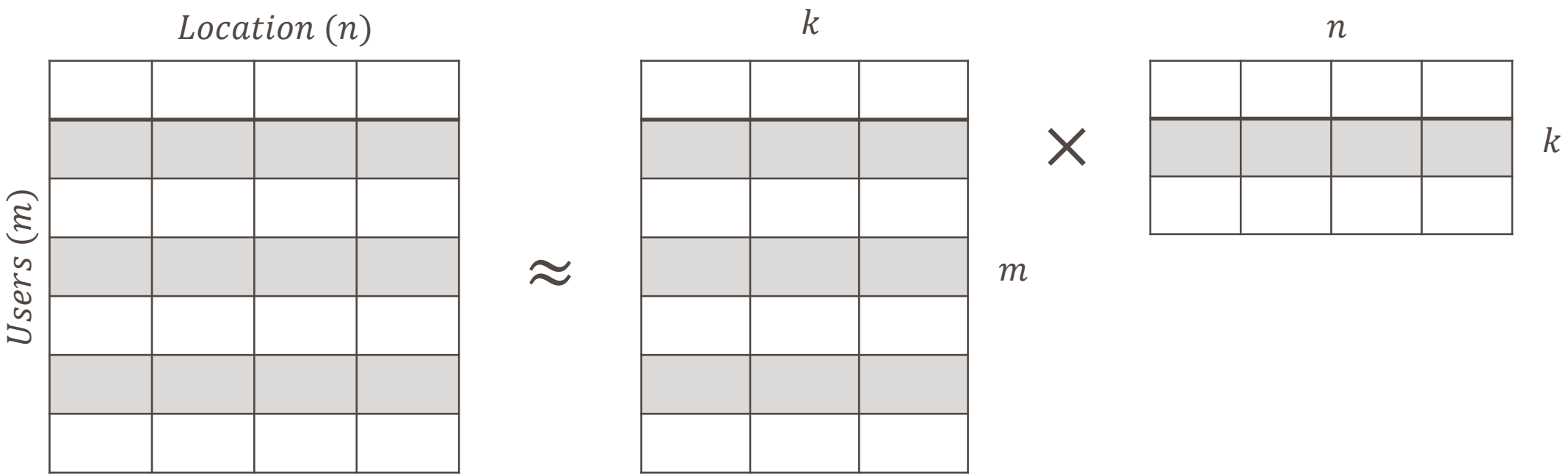


Recommendation Methods

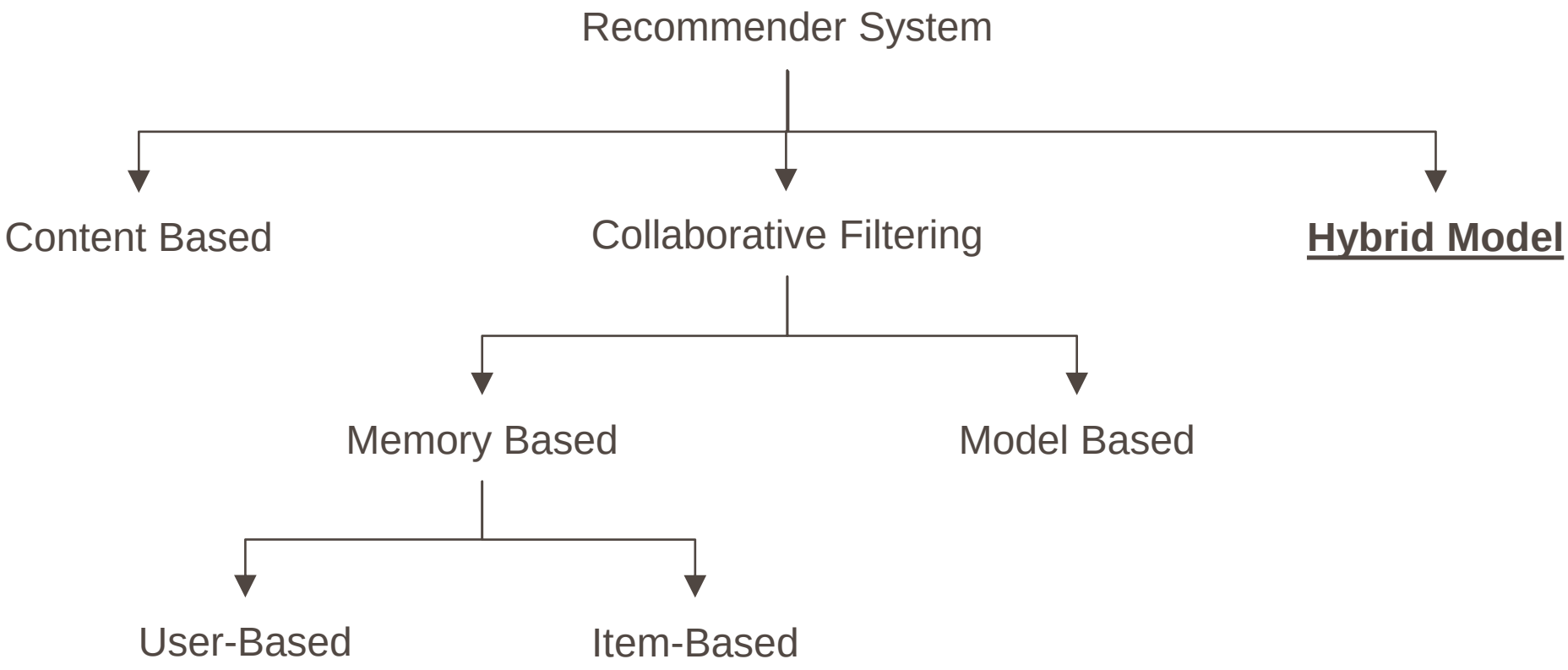


Matrix Factorization

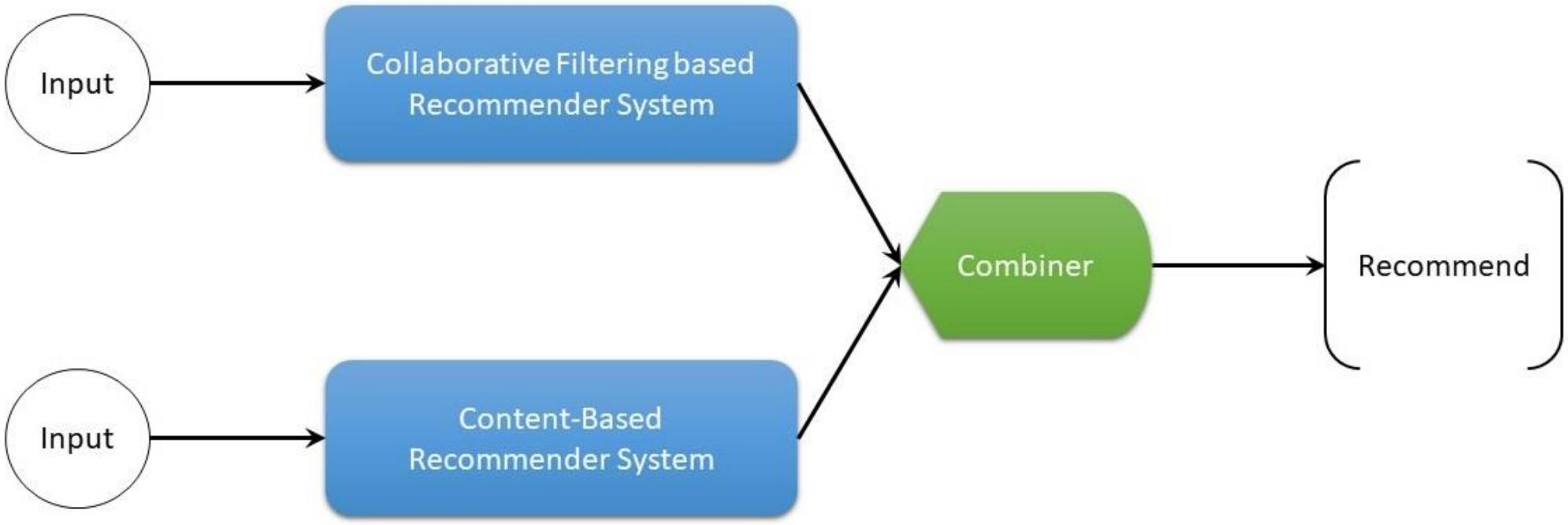
- Latent Features (k)



Recommendation Methods

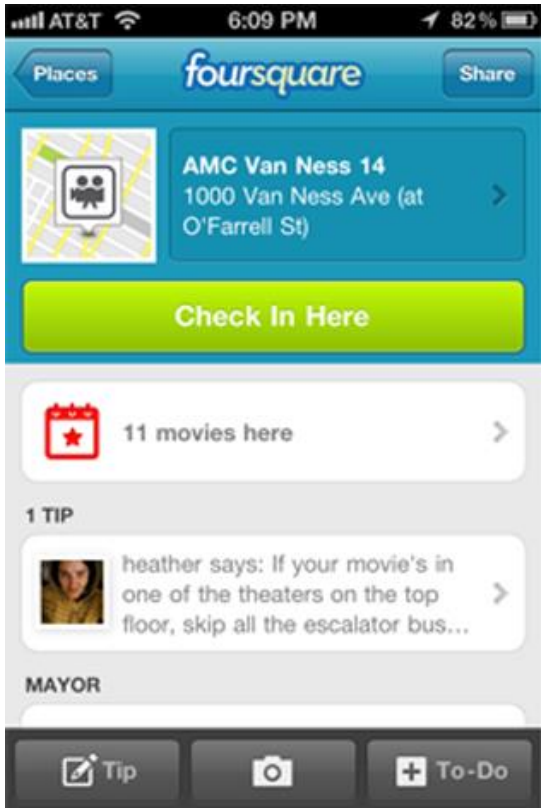


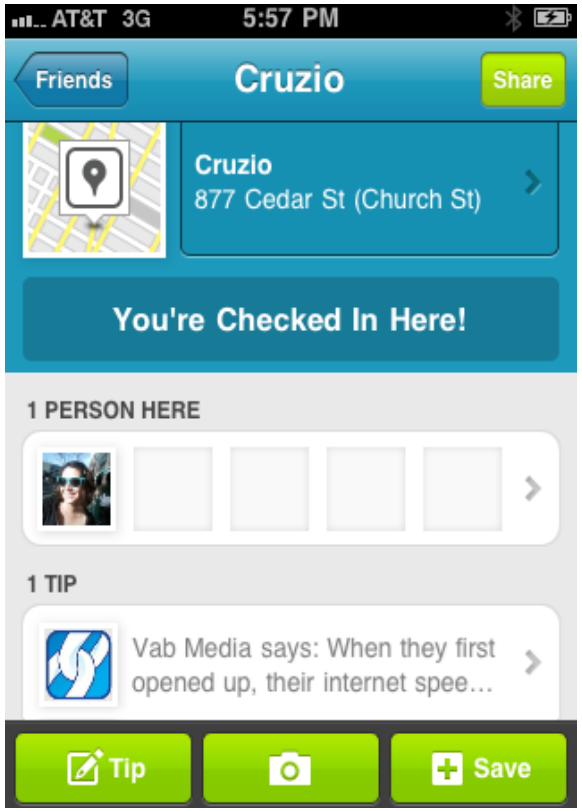
Hybrid Model



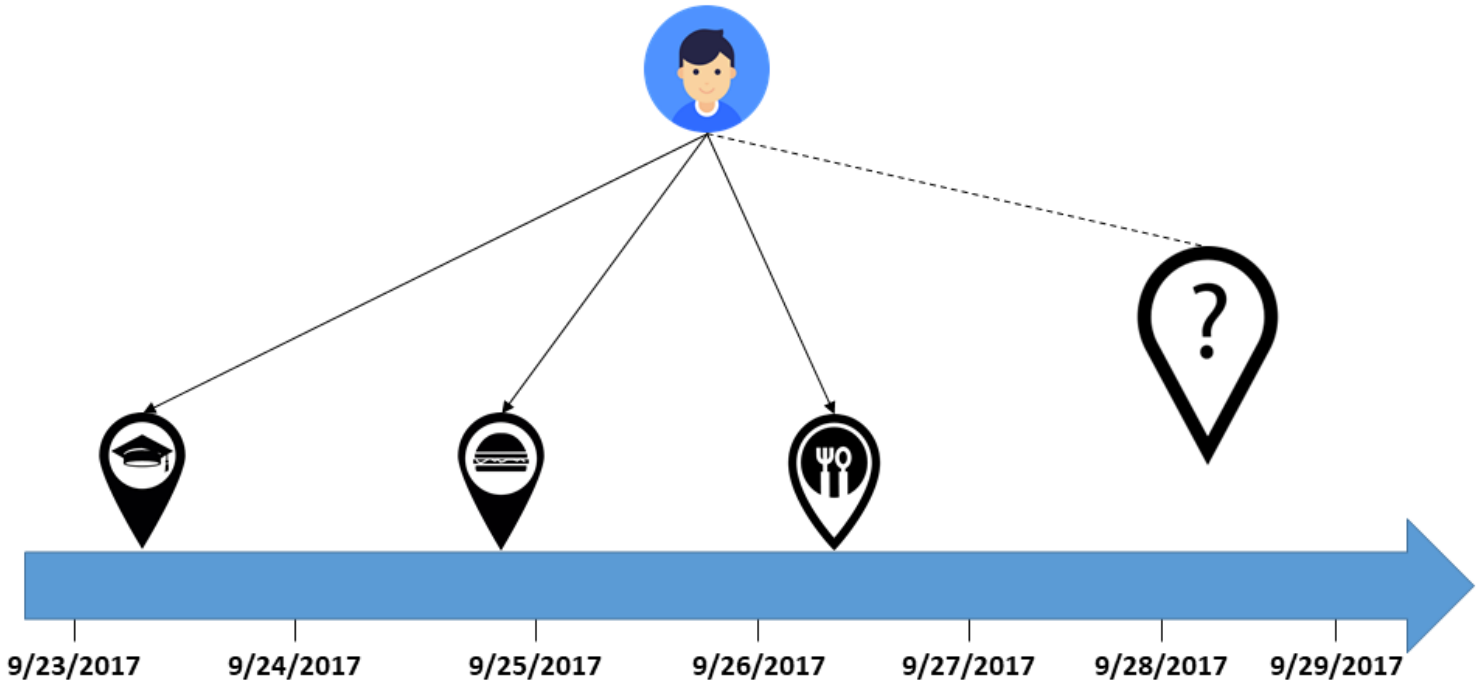
Check-ins

- Check-in becomes a Life Style





Point-of-Interest Recommendation



Challenges

- Data Sparsity
  - Low percentage of rated (*checked-in*) items (*locations*)
  - *Foursquare: 50 million user, 105 million venues, 12 billion check-ins*
- Scalability
  - The number of users and items
- Cold-Start
  - New user or item
  - Little information

Challenges (Cont.)

- User Feedback
  - Explicit Feedback Data
    - *Rating or like and dislike*
  - Implicit Feedback Data
    - *User visited or bought object*

Check-in Information

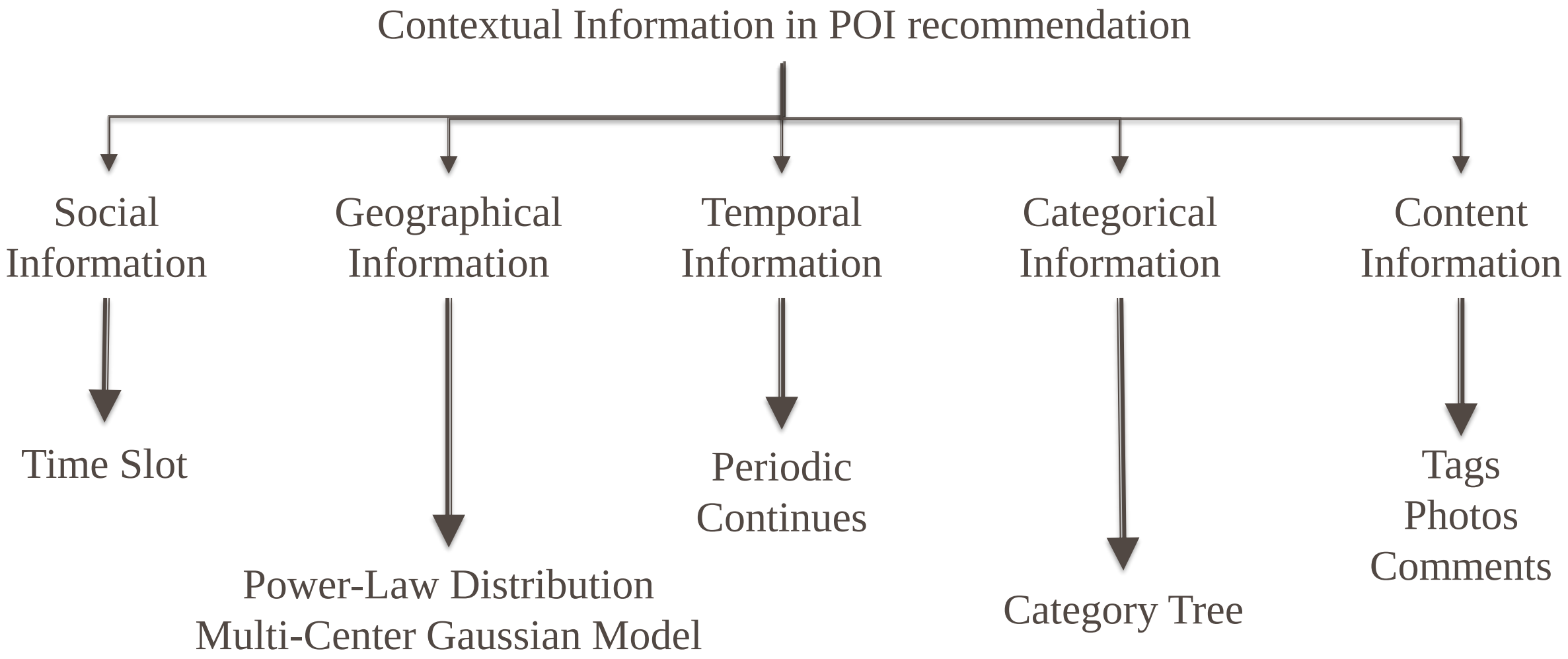
Location

Time

Comments

Friends

Contextual Information



Contextual Information in Related Works

Table 1. Summary of contextual information in related works

Related Work	Geographical	Temporal	Social	Category	Content
Xin Liu 2013 [29]				✓	
Jean-Benoit Griesner 2015 [23]		✓			
Ramesh Baral 2016 [30]	✓	✓	✓	✓	
Ramesh Baral 2016 [31]	✓	✓	✓	✓	
Xutao Li 2015 [19]	✓	✓			
Shenglin Zhao 2016 [24]		✓			
Chen Cheng 2016 [20]	✓				
Chen Cheng 2012 [10]	✓		✓		
Zhiyuan Zhang 2018 [11]	✓		✓		✓
Shaojie Qiao 2018 [27]			✓		
Mohammadreza Rahimi 2017 [25]		✓	✓		
Saeid Hosseini 2016 [26]		✓			

References

1. Y. Zheng, "Trajectory Data Mining: An Overview," ACM Transactions on Intelligent Systems and Technology (TIST), vol. 6, no. 3, p. 29, 2015.

2. J. D. Mazimpaka and S. Timpf, "Trajectory Data Mining: A Review of Methods and Applications," Journal of Spatial Information Science, no. 13, pp. 61-99, 2016.

3. J. Bao, Y. Zheng, D. Wilkie and . M. Mokbel, "Recommendations in Location-Based Social Networks: A Survey," GeoInformatica, vol. 19, no. 3, pp. 525-565, 2015.

4. Z. Ding, X. Li, C. Jiang and M. Zhou, "Objectives and State-of-the-Art of Location-Based Social Network Recommender Systems," ACM Computing Surveys (CSUR), vol. 51, no. 1, p. 18, 2018.

5. Y. Yu and X. Chen, A Survey of Point-of-Interest Recommendation in Location-Based Social Networks, Workshops at the Twenty-Ninth AAAI Conference on Artificial Intelligence, 2015.

6. J.-B. Griesner, T. Abdessalem and H. Naacke, "POI Recommendation: Towards Fused Matrix Factorization with Geographical and Temporal Influences," in Proceedings of the 9th ACM Conference on Recommender Systems, 2015.

References

7. S. Zhao, I. King and M. R. Lyu, "A Survey of Point-of-Interest Recommendation in Location-Based Social Networks," arXiv preprint arXiv:1607.00647, 2016.

8. X. Li, G. Cong, X. Li, T.-A. Nguyen Pham and S. Krishnaswamy, "Rank-GeoFM: A Ranking based Geographical Factorization Method for Point of Interest Recommendation," in Proceedings of the 38th International ACM SIGIR Conference on Research and Development in Information Retrieval, Santiago, 2015.

9. C. Cheng, H. Yang, I. King and M. R. Lyu, "A Unified Point-of-Interest Recommendation Framework in Location-Based Social Networks," ACM Transactions on Intelligent Systems and Technology (TIST, vol. 8, no. 1, p. 10, 2016.

10. Y. Liu, T.-A. N. Pham, G. Cong and Q. Yuan, "An Experimental Evaluation of Point-of-Interest Recommendation in Location-Based Social Networks," VLDB, vol. 10, no. 10, pp. 1010-1021, 2017.

Thanks!

Questions or Comments?